

● if Statement (Q1–Q4)

- Q1.** Accept a number from the user. If it is positive, print "Positive Number".
- Q2.** Accept a year from the user. If it is divisible by 4, print "Possibly a Leap Year".
- Q3.** Accept marks. If marks are greater than or equal to 90, print "Excellent Performance".
- Q4.** Accept a person's age. If age is greater than or equal to 60, print "Senior Citizen".
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● if-else Statement (Q5–Q9)

- Q5.** Accept a number and check whether it is even or odd using if-else.
- Q6.** Accept two numbers and print which one is greater (or if they are equal).
- Q7.** Accept a person's age. Use if-else to print "Minor" or "Adult" (18+).
- Q8.** Accept a number. Print "Positive" if greater than 0, otherwise print "Non-Positive".
- Q9.** Accept the price of an item. If price is above ₹1000, print "Premium Product", otherwise print "Standard Product".
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● Nested if Statement (Q10–Q14)

- Q10.** Accept age and citizenship status (boolean). Using nested if, print "Eligible to Vote" only if age ≥ 18 AND the person is a citizen.
- Q11.** Accept three numbers. Using nested if statements (no logical operators), find and print the largest of the three.
- Q12.** A bank loan is approved only if income > 50000 AND credit score > 700 . Use nested if to check both conditions and print the result.
- Q13.** Accept marks and attendance percentage. Using nested if, print "Eligible for Exam" only if marks ≥ 35 AND attendance ≥ 75 .
- Q14.** Accept a username and password. Using nested if, check if username equals "admin" first, then check if password equals "1234" inside it. Print "Login Successful" or appropriate failure messages at each level.
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● switch Statement (Q15–Q18)

Q15. Accept a number from 1–12 and use switch to print the corresponding month name.

Q16. Accept a character grade ('A', 'B', 'C', 'D', 'F') and use switch to print the remark ("Excellent", "Good", "Average", "Below Average", "Fail").

Q17. Accept an operator (+, -, *, /) and two numbers. Use switch on the operator to perform the corresponding arithmetic operation and print the result.

Q18. Accept a number representing a day (1–7) and use switch to print whether it's a "Weekday" or "Weekend" (group cases 1–5 together using fall-through, and 6–7 together).

● **Mixed / Challenge (Q19–Q20)**

Q19. Traffic Light Simulator — Accept a signal color as a String ("Red", "Yellow", "Green") using switch. If the signal is "Green", use a nested if to also check a boolean pedestrianWaiting — if true, print "Stop briefly for pedestrian", else print "Go".

Q20. Simple Calculator Menu — Display a menu using println (1-Add, 2-Subtract, 3-Multiply, 4-Divide). Accept the user's choice and two numbers. Use switch to perform the operation. Inside the Divide case, use a nested if to check if the second number is 0 — if so print "Cannot divide by zero", otherwise print the result.